

## SMAS-1 | Assignment 3: Image Transformations

### **A1 = [[2,0],[0,0.5]] (Scaling)**

(a)  $T(e_1) = [2, 0]^T$ ,  $T(e_2) = [0, 0.5]^T$

(c) Rank = 2

(d) No dimension/information is lost.

(e) The image stretches to twice its original width (first column scales  $x$  by 2) and compresses to half its height (second column scales  $y$  by 0.5).

### **A2 = [[0,-1],[1,0]] (90 Degree Rotation)**

(a)  $T(e_1) = [0, 1]^T$ ,  $T(e_2) = [-1, 0]^T$

(c) Rank = 2

(d) No dimension/information is lost.

(e) The  $x$ -axis maps to the positive  $y$ -axis and the  $y$ -axis maps to the negative  $x$ -axis, visually rotating the image 90 degrees counter-clockwise.

### **A3 = [[1,1],[0,1]] (Horizontal Shear)**

(a)  $T(e_1) = [1, 0]^T$ ,  $T(e_2) = [1, 1]^T$

(c) Rank = 2

(d) No dimension/information is lost.

(e) The  $x$ -axis points remain unchanged (first column), while the  $y$ -axis points are pushed to the right proportionally to their height (second column), slanting the image.

### **A4 = [[-1,0],[0,1]] (Reflection in $y$ -axis)**

(a)  $T(e_1) = [-1, 0]^T$ ,  $T(e_2) = [0, 1]^T$

(c) Rank = 2

(d) No dimension/information is lost.

(e) The  $x$ -coordinates are mapped to their opposite signs (first column is negative) while  $y$ -coordinates remain the same, acting as a mirror across the vertical center.

### **A5 = [[1,0],[0,0]] (Projection onto $x$ -axis)**

(a)  $T(e_1) = [1, 0]^T$ ,  $T(e_2) = [0, 0]^T$

(c) Rank = 1

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(d) Dimension/information IS lost. The 2D image collapses into a 1D space.

(e) Because the second column is entirely zeros, every pixel's vertical position is forced to zero, collapsing the image into a single flat horizontal line.